



```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("✅ Libraries loaded successfully!")
```

✅ Libraries loaded successfully!

```
In [1]: import os
print(os.listdir())

['.anaconda', '.conda', '.condarc', '.continuum', '.ipynb_checkpoints', '.ipython', '.jupyter', '.matplotlib', '.vscode', 'anaconda3', 'AppData', 'Application Data', 'Contacts', 'Cookies', 'Documents', 'Downloads', 'Favorites', 'learn flu', 'Links', 'Local Settings', 'Music', 'My Documents', 'NetHood', 'NTUSER.DAT', 'ntuser.dat.LOG1', 'ntuser.dat.LOG2', 'NTUSER.DAT{1bf12db1-b527-11f0-8135-a059503b2b9d}.TM.blf', 'NTUSER.DAT{1bf12db1-b527-11f0-8135-a059503b2b9d}.TMContainer00000000000000000001.regtrans-ms', 'NTUSER.DAT{1bf12db1-b527-11f0-8135-a059503b2b9d}.TMContainer00000000000000000002.regtrans-ms', 'ntuser.ini', 'OneDrive', 'PrintHood', 'Recent', 'Saved Games', 'Searches', 'SendTo', 'Start Menu', 'supply_chain_project.ipynb', 'Templates', 'Untitled.ipynb', 'Videos']
```

```
In [4]: import pandas as pd

data = pd.read_csv("supply_chain_data.csv")
data.head()
```

```
Out[4]:
```

	Product type	SKU	Price	Availability	Number of products sold	Revenue generated	Customer demographics
0	haircare	SKU0	69.808006	55	802	8661.996792	Non-binary
1	skincare	SKU1	14.843523	95	736	7460.900065	Female
2	haircare	SKU2	11.319683	34	8	9577.749626	Unknown
3	skincare	SKU3	61.163343	68	83	7766.836426	Non-binary
4	skincare	SKU4	4.805496	26	871	2686.505152	Non-binary

5 rows × 24 columns

```
In [5]: # Basic info about dataset
data.info()

# Check missing values
data.isnull().sum()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Product type                          100 non-null    object
1   SKU                                   100 non-null    object
2   Price                                100 non-null    float64
3   Availability                          100 non-null    int64
4   Number of products sold              100 non-null    int64
5   Revenue generated                    100 non-null    float64
6   Customer demographics                100 non-null    object
7   Stock levels                         100 non-null    int64
8   Lead times                           100 non-null    int64
9   Order quantities                     100 non-null    int64
10  Shipping times                       100 non-null    int64
11  Shipping carriers                    100 non-null    object
12  Shipping costs                       100 non-null    float64
13  Supplier name                       100 non-null    object
14  Location                             100 non-null    object
15  Lead time                            100 non-null    int64
16  Production volumes                   100 non-null    int64
17  Manufacturing lead time              100 non-null    int64
18  Manufacturing costs                  100 non-null    float64
19  Inspection results                   100 non-null    object
20  Defect rates                         100 non-null    float64
21  Transportation modes                 100 non-null    object
22  Routes                              100 non-null    object
23  Costs                               100 non-null    float64
dtypes: float64(6), int64(9), object(9)
memory usage: 18.9+ KB

```

```
Out[5]: Product type      0
        SKU              0
        Price            0
        Availability      0
        Number of products sold 0
        Revenue generated 0
        Customer demographics 0
        Stock levels      0
        Lead times        0
        Order quantities  0
        Shipping times    0
        Shipping carriers 0
        Shipping costs    0
        Supplier name     0
        Location          0
        Lead time         0
        Production volumes 0
        Manufacturing lead time 0
        Manufacturing costs 0
        Inspection results 0
        Defect rates      0
        Transportation modes 0
        Routes            0
        Costs             0
        dtype: int64
```

```
In [6]: data = data.dropna()
        data.isnull().sum()
```

```
Out[6]: Product type      0
        SKU              0
        Price            0
        Availability      0
        Number of products sold 0
        Revenue generated 0
        Customer demographics 0
        Stock levels      0
        Lead times        0
        Order quantities  0
        Shipping times    0
        Shipping carriers 0
        Shipping costs    0
        Supplier name     0
        Location          0
        Lead time         0
        Production volumes 0
        Manufacturing lead time 0
        Manufacturing costs 0
        Inspection results 0
        Defect rates      0
        Transportation modes 0
        Routes            0
        Costs             0
        dtype: int64
```

```
In [7]: # Check first few rows again
data.head()

# Check numeric summaries
data.describe()

# Check all unique categories in each column
for col in data.select_dtypes(include='object').columns:
    print(f"{col}: {data[col].nunique()} unique values")
```

```
Product type: 3 unique values
SKU: 100 unique values
Customer demographics: 4 unique values
Shipping carriers: 3 unique values
Supplier name: 5 unique values
Location: 5 unique values
Inspection results: 3 unique values
Transportation modes: 4 unique values
Routes: 3 unique values
```

```
In [8]: data.groupby('Product type')['Revenue generated'].sum().sort_values(ascending=
```

```
Out[8]: Product type
skincare      241628.162133
hairecare     174455.390605
cosmetics     161521.265999
Name: Revenue generated, dtype: float64
```

```
In [9]: data.groupby('Supplier name')['Lead time'].mean().sort_values()
```

```
Out[9]: Supplier name
Supplier 1    14.777778
Supplier 4    15.222222
Supplier 5    18.055556
Supplier 2    18.545455
Supplier 3    20.133333
Name: Lead time, dtype: float64
```

```
In [11]: data.columns
```

```
Out[11]: Index(['Product type', 'SKU', 'Price', 'Availability',
               'Number of products sold', 'Revenue generated', 'Customer demographic
               s',
               'Stock levels', 'Lead times', 'Order quantities', 'Shipping times',
               'Shipping carriers', 'Shipping costs', 'Supplier name', 'Location',
               'Lead time', 'Production volumes', 'Manufacturing lead time',
               'Manufacturing costs', 'Inspection results', 'Defect rates',
               'Transportation modes', 'Routes', 'Costs'],
              dtype='object')
```

```
In [14]: data['Profit'] = data['Revenue generated'] - (data['Manufacturing costs'] + da
data.head()
```

Out[14]:

	Product type	SKU	Price	Availability	Number of products sold	Revenue generated	Customer demographics
0	haircare	SKU0	69.808006	55	802	8661.996792	Non-binary
1	skincare	SKU1	14.843523	95	736	7460.900065	Female
2	haircare	SKU2	11.319683	34	8	9577.749626	Unknown
3	skincare	SKU3	61.163343	68	83	7766.836426	Non-binary
4	skincare	SKU4	4.805496	26	871	2686.505152	Non-binary

5 rows × 25 columns

```
In [15]: data[['Product type', 'Revenue generated', 'Manufacturing costs', 'Costs', 'Pr
```

Out[15]:

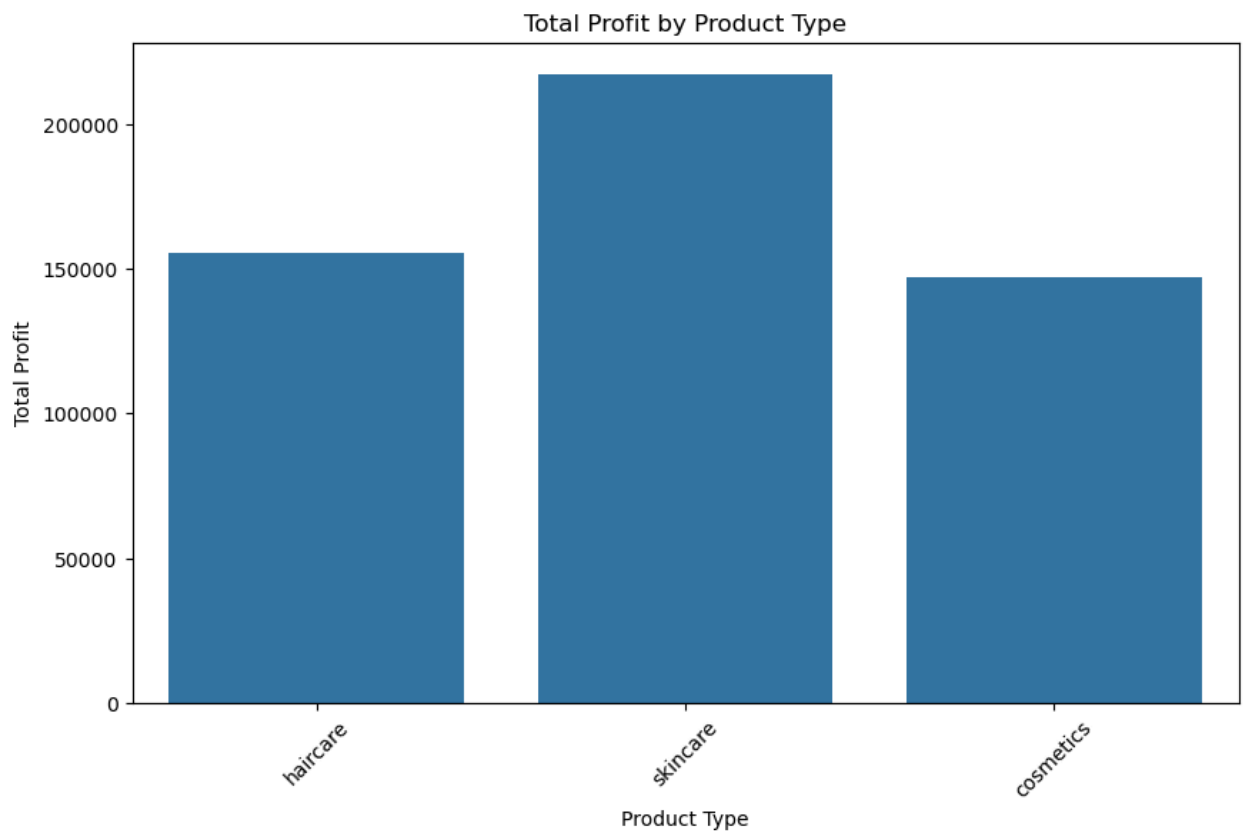
	Product type	Revenue generated	Manufacturing costs	Costs	Profit
0	haircare	8661.996792	46.279879	187.752075	8427.964838
1	skincare	7460.900065	33.616769	503.065579	6924.217717
2	haircare	9577.749626	30.688019	141.920282	9405.141325
3	skincare	7766.836426	35.624741	254.776159	7476.435525
4	skincare	2686.505152	92.065161	923.440632	1670.999359

```
In [16]: data.groupby('Product type')['Profit'].sum().sort_values(ascending=False)
```

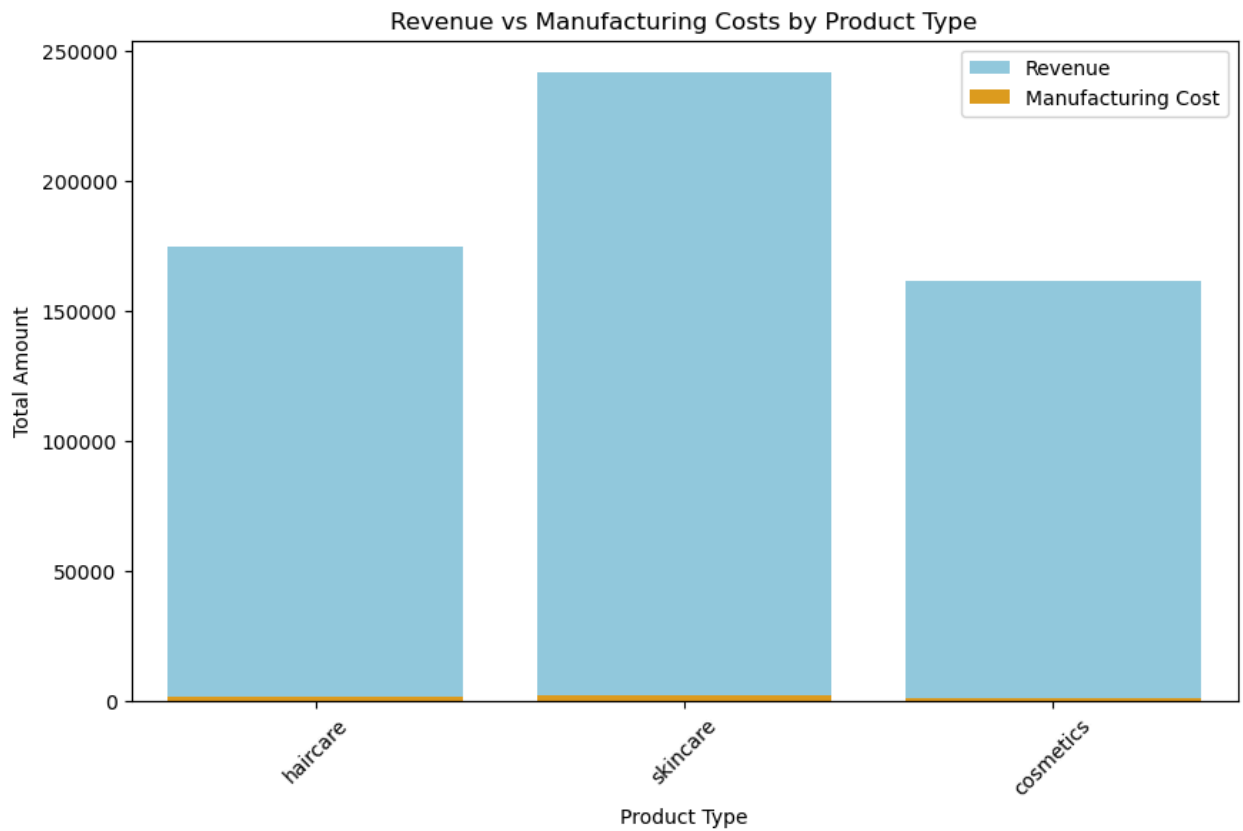
```
Out[16]: Product type
skincare    217439.117770
haircare    155478.955964
cosmetics   147035.497463
Name: Profit, dtype: float64
```

```
In [18]: import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,6))
sns.barplot(x='Product type', y='Profit', data=data, estimator=sum, errorbar=None)
plt.title('Total Profit by Product Type')
plt.xlabel('Product Type')
plt.ylabel('Total Profit')
plt.xticks(rotation=45)
plt.show()
```



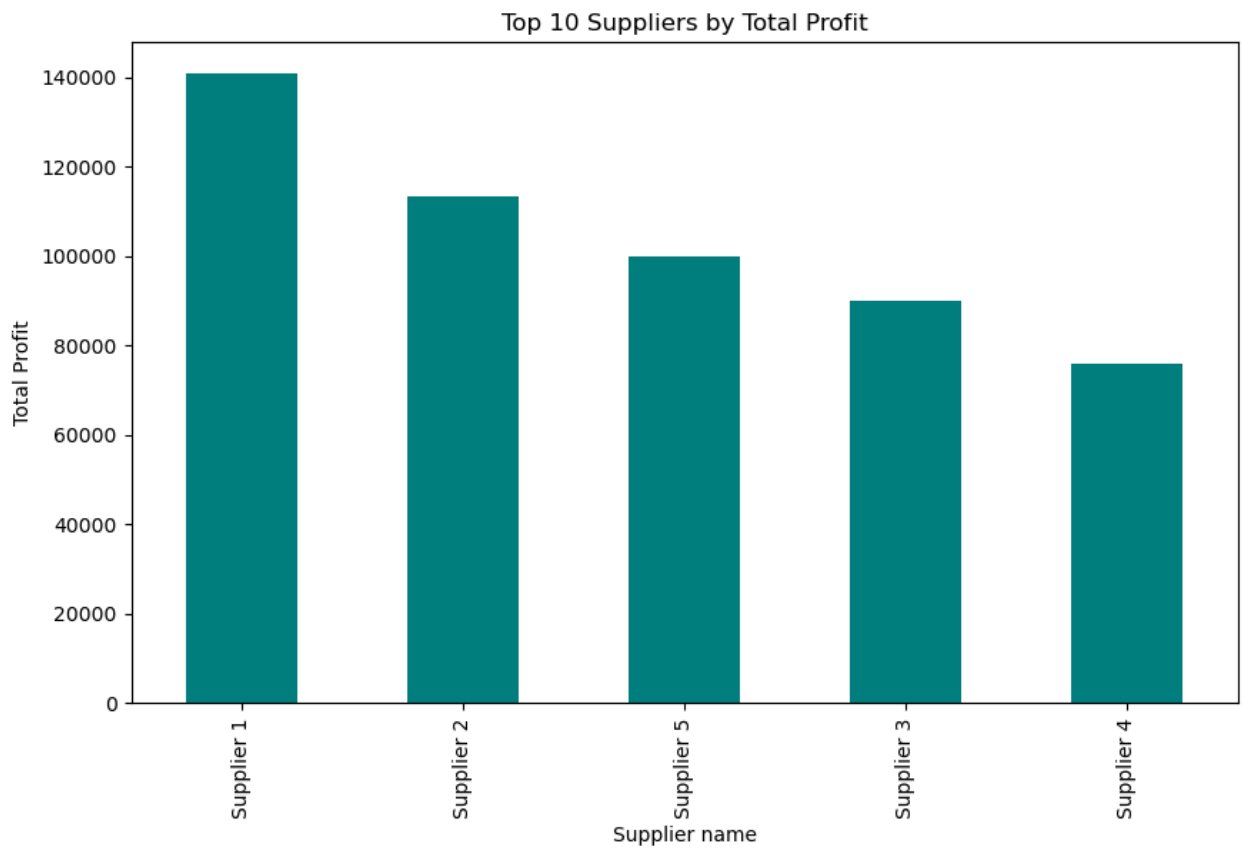
```
In [19]: plt.figure(figsize=(10,6))
sns.barplot(x='Product type', y='Revenue generated', data=data, estimator=sum,
sns.barplot(x='Product type', y='Manufacturing costs', data=data, estimator=su
plt.legend()
plt.title('Revenue vs Manufacturing Costs by Product Type')
plt.xlabel('Product Type')
plt.ylabel('Total Amount')
plt.xticks(rotation=45)
plt.show()
```



```
In [20]: supplier_profit = data.groupby('Supplier name')['Profit'].sum().sort_values(ascending=False)
print(supplier_profit.head(10))
```

```
Supplier name
Supplier 1    140786.155560
Supplier 2    113221.123995
Supplier 5     99889.226150
Supplier 3     90109.466295
Supplier 4     75947.599198
Name: Profit, dtype: float64
```

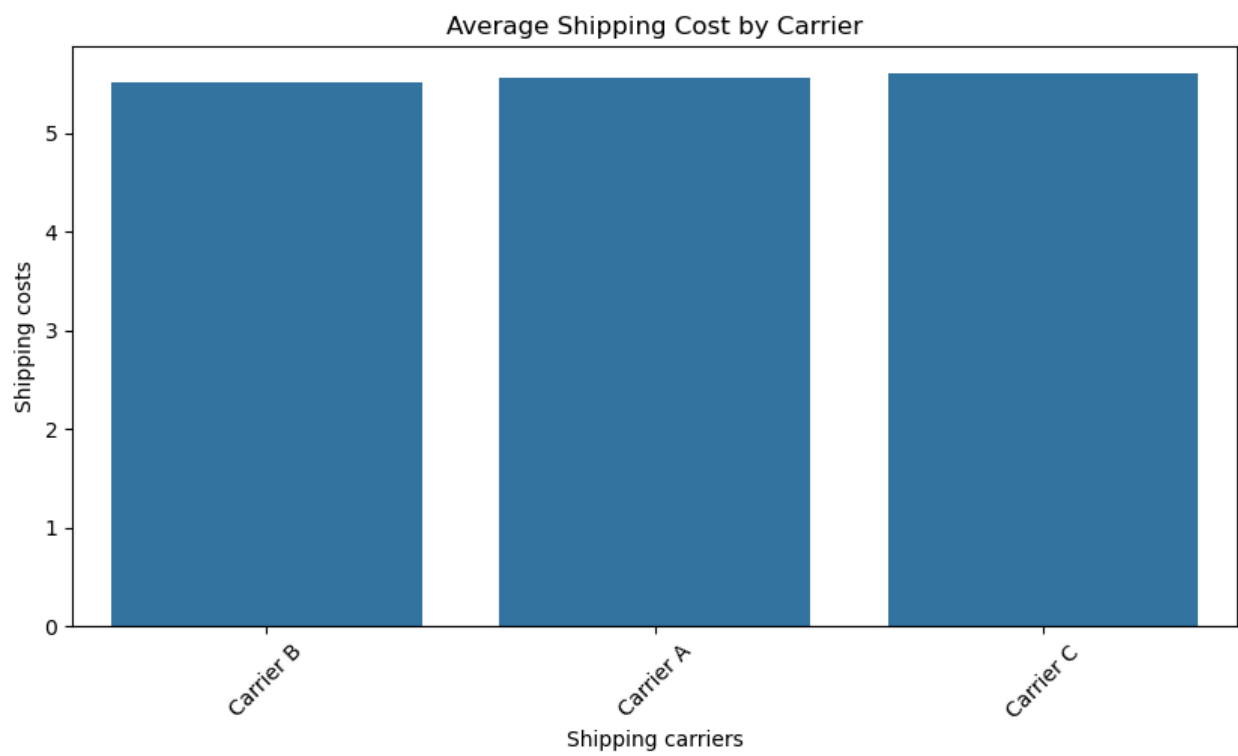
```
In [21]: plt.figure(figsize=(10,6))
supplier_profit.head(10).plot(kind='bar', color='teal')
plt.title('Top 10 Suppliers by Total Profit')
plt.ylabel('Total Profit')
plt.show()
```



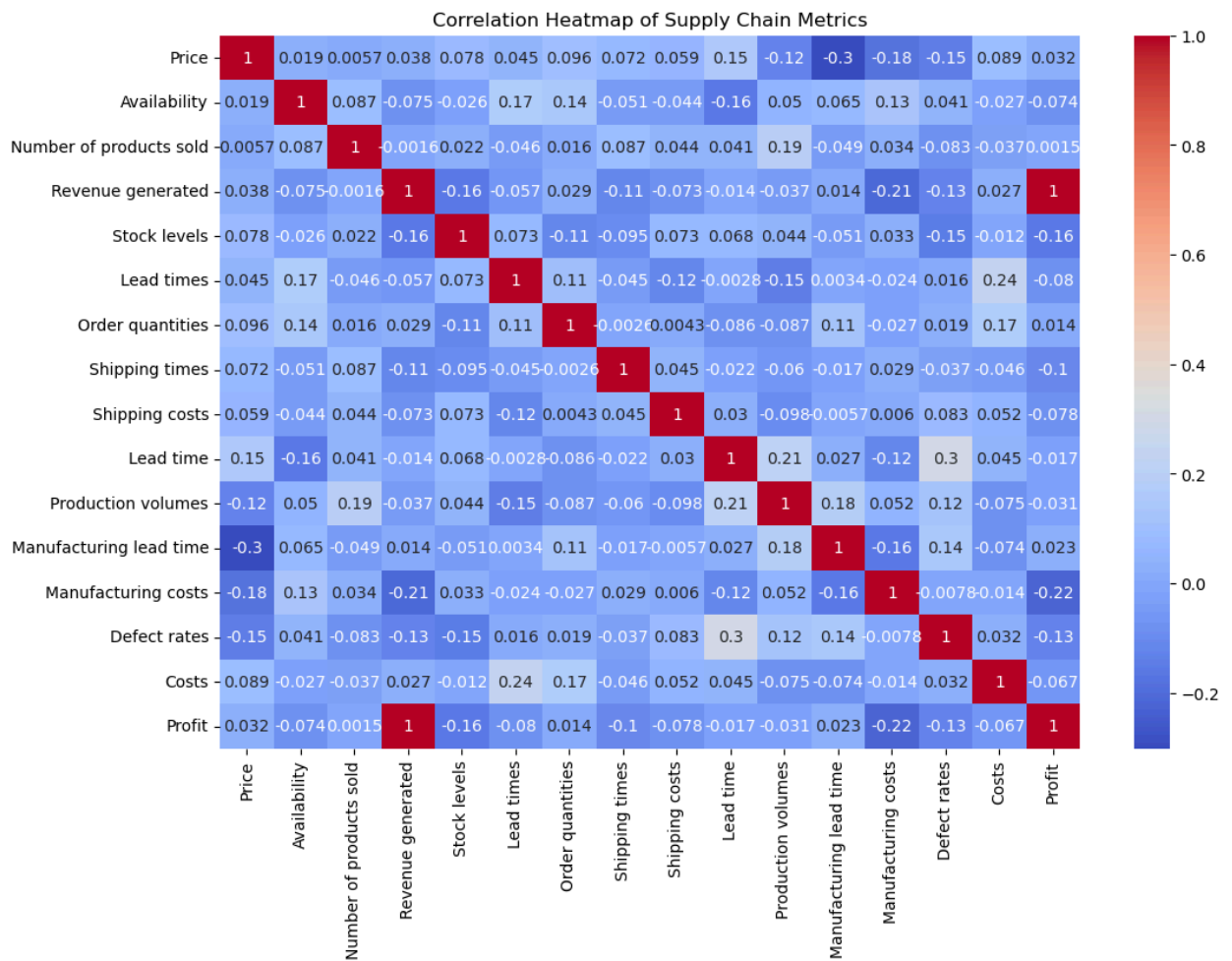
```
In [22]: data.groupby('Shipping carriers')['Shipping costs'].mean().sort_values(ascending=True)
```

```
Out[22]: Shipping carriers
Carrier C    5.599292
Carrier A    5.554923
Carrier B    5.509247
Name: Shipping costs, dtype: float64
```

```
In [23]: plt.figure(figsize=(10,5))
sns.barplot(x='Shipping carriers', y='Shipping costs', data=data, estimator='mean')
plt.title('Average Shipping Cost by Carrier')
plt.xticks(rotation=45)
plt.show()
```

```
In [24]: plt.figure(figsize=(12,8))
sns.heatmap(data.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap of Supply Chain Metrics')
plt.show()
```



```
In [25]: data.to_csv("final_supply_chain_analysis.csv", index=False)
print("✅ Final analyzed dataset saved successfully!")
```

✅ Final analyzed dataset saved successfully!

In []: